MIL-G-0015427A(DM)
5 December 1967
USED IN LIEU OF
MIL-W-15427(BuMed)
15 July 1950

MILITARY SPECIFICATION

GLASS WOOL, FILTERING

This limited coordination Military specification has been prepared by Defense Personnel Support Center based upon currently available technical information but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-W-15427(BuMed). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers glass wool suitable for use as a filtering medium.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6640

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STANDARDS

Military

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The glass wool shall be fabricated from borosilicate glass having a thermal coefficient of expansion not exceeding 0.0000075 cm./cm./1° C. The glass shall be resilient, soft, and pliable.

3.2 Impurities.

3.2.1 Alkalinity. The alkali content shall not exceed the equivalent of 0.45 percent Na_2O, when tested as specified in 4.4.1.

3.2.2 Heavy metals. The glass wool shall contain no heavy metals, when tested as specified in 4.4.2.

3.3 Fiber diameter. The average fiber diameter shall be not less than 0.00025 inch or more than 0.00050 inch. Diameter shall be determined by in-process testing.

3.4 Size and unit. The glass wool shall be furnished in sheets, and shall be wrapped in rolls of the size and unit specified.

3.5 Workmanship. Workmanship shall be first class throughout. The glass wool shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to insure that supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as shown in table I. The unit of product shall be one roll.

Table I. Tests

Characteristic	Requirement	Test procedure	Inspection level	AQL
Alkalinity	3.2.1	4.4.1	S-1	*
Heavy metals	3.2.2	4.4.2	S-1	*

* Acceptance number shall be zero.

4.3 Examination. The rolls of glass wool shall be examined to determine compliance with all requirements contained in this specification.

4.4 Tests.

4.4.1 Alkalinity. Exactly one gram of wool shall be introduced into a 250 milliliter (ml.) low alkali borosilicate glass Erlenmeyer flask fitted with a one-hole rubber stopper. One-hundred ml. of distilled water at room temperature shall be added to the flask. After situating the flask under a low alkali borosilicate glass reflux condenser, the water shall be brought to a boil and boiled for 4 hours. The condenser shall then be rinsed down with a few ml. of distilled water, and the flask shall be immediately cooled in running tap water to room temperature. Six drops of phenol red indicator shall be added, and the solution shall be titrated with standardized N/50 H_2SO_4 to the complete disappearance of the red color. With each series of determinations, a blank shall be run using 100 ml. of distilled water.

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4.4.2 Absence of heavy metals. Exactly one gram of wool shall be introduced into a 250 ml. low alkali borosilicate glass Erlenmeyer flask. One hundred ml. of approximately 0.2N HCl shall be added to the flask. The flask shall be fitted with a one-hole stopper equipped with a piece of low alkali borosilicate glass tubing extending approximately 12 inches above the flask. The flask shall be immersed to a depth of about 2 inches in a thermostatically controlled water bath held at 9.0. $\pm$ 0.5° C. After 4 hours the flask shall be removed from the water bath and cooled immediately to room temperature in running tap water. The solution shall be filtered, and the wool shall be washed with three 25 ml. portions of distilled water. H_2S gas shall be bubbled through the filtrate which has been warmed to approximately 70° C., for 1/2 hour, and the solution shall be compared to a blank which has been carried along with the sample through the entire procedure. Equal clarity of blank and sample shall be considered evidence that there are not heavy metals in the glass wool.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One roll (1 lb.), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	Not applicable	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Each roll shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors shall be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4, of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, constructed so as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.2).

6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
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